

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Examination (Supplementary) – 2023

Course: B. Tech. Branch: Electronics & Computer Engg./Electronics & Computer Science Engg.

Semester: III

**Subject Code & Name: (BTECPC303) Programming, Data Structure & Algorithm
Using C**

Max Marks: 60

Date:14/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
-

	Marks
Q.1: Solve Any Two of the following.	12
A. Define Data Structure & Explain its types.	6
B. Explain different operations on Data structure.	6
C. What is Linked List & types?	6
Q.2: Solve Any Two of the following.	12
A. Explain Stack & it's representation.	6
B. What are the operations of Stack?	6
C. Applications of Stack.	6
Q.3: Solve Any Two of the following.	12
A. Describe Queue & it's types.	6
B. Explain Queue Operations.	6
C. Define tree and its terminologies.	6
Q.4: Solve Any Two of the following.	12
A. Explain Binary Tree traversal.	6
B. Write Binary Tree Properties.	6
C. Define Graph & it's Five Types.	6
Q.5: Solve Any Two of the following.	12
A. Explain Adjacency Matrix.	6
B. Describe Algorithm Analysis.	6
C. Explain Algorithm Complexity.	6